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Quantum Infused Multi-Modal Sentiment Analysis of Low Resourced Language

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***Abstract*—For Urdu sentiment analysis, we present a quantum-enhanced multimodal system that combines a variational quantum circuit (VQC) classifier with traditional feature extractors. To create a cohesive embedding, a cross-modal transformer model first processes raw text, audio, and visual information. On a noisy intermediate-scale quantum (NISQ) processor, these fused embeddings are then angle-encoded into a parameterized quantum state. Simultaneous perturbation stochastic approximation (SPSA) is used to improve the trainable rotation gates of the VQC under realistic hardware-inspired noise models. Our quantum-enhanced model outperforms the classical cross-modal baseline (82.4%/79.6%) and a purely classical MLP head (83.1%/80.8%) with an overall accuracy of 84.2% and a macro-F1 score of 81.7% when tested on UMSA-U, a curated dataset of 6,000 Urdu video clips with aligned transcripts, audio, and facial expression annotations. The findings show that noise-robust SPSA tuning maintains performance on simulated hardware error rates up to 5%, while quantum feature encoding produces more separable decision boundaries. These findings establish a new standard for Urdu sentiment recognition on NISQ devices and demonstrate the potential of variational quantum classifiers for low-resource, multimodal language solutions.**

***Index Terms*—Quantum Machine Learning, Variational Circuits, Urdu NLP, Multimodal Sentiment Analysis**

I. Introduction

Sentiment analysis has been a fundamental topic of natural language processing, providing insights on public opinion, customer opinions and social media postings [1] [2]. Progressing with these approaches for the high-resource languages like English and Mandarin, deploying them to low-resource languages is still a challenge because of less available annotated corpora and tool support. With more than 230 million speakers across the globe [3], Urdu is representative of this divide: a highly morphologically rich language written in an Arabic script, with limited sentiment annotated data and less developed linguistic resources. Multimodal sentiment analysis aims to mitigate the deficiencies of text-only methods by incorporating the complementary cues provided by the audio and visual streams [4], [5]. Prosodic cues in speech and facial expressions frequently contain emotional nuances that are lost when relying on text alone, particularly in informal, contextually-laden content such as video blogs or customer testimonials. However, most of the existing multimodal architectures are specifically designed for major languages with no available Urdu data and experienced enhancement under real-world noise.

Quantum machine learning [6], [7] has emerged as a promising avenue for pattern recognition, offering potential advantages in feature representation and optimization efficiency. Variational quantum circuits (VQCs) [8], in particular, apply parameterized quantum gates to encode high-dimensional embeddings and exploit quantum interference for classification tasks. When trained via noise-sturdy optimizers such as simultaneous perturbation stochastic approximation (SPSA) [9], VQCs can tolerate the error rates inherent to noisy intermediate-scale quantum (NISQ) devices [10]. To date, quantum methods have largely focused on benchmark datasets in controlled environments, with little attention to multimodal language applications or low-resource settings.

In this study, we introduce a quantum-enhanced multimodal sentiment analysis pipeline specifically designed for Urdu, bridging these domains. To create a unified embedding, raw text, audio, and visual characteristics are first combined using a cross-modal transformer. After that, this embedding is angle-encoded into a VQC, and SPSA is used to improve its parameters under actual noise models. Using UMSA-U, a new dataset of 6,000 Urdu video clips tagged in text, audio, and

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visual modalities, we assess our method. Our key contributions are as follows:

- We curate UMSA-U, the first large-scale multimodal Urdu sentiment dataset with synchronized transcripts, audio cues, and facial expression annotations.
- We design a cross-modal transformer that effectively fuses heterogeneous features into a compact representation suitable for quantum encoding.
- We integrate a VQC classifier trained with noise-aware SPSA, demonstrating robustness to hardware-inspired perturbations and competitive accuracy gains.
- We establish new benchmarks on UMSA-U, achieving 84.2% accuracy and a macro-F1 score of 81.7%, surpassing classical baselines and highlighting the benefits of quantum feature encoding.

Through comprehensive ablations and noise-robustness studies, we provide practical guidelines for deploying variational algorithms on NISQ hardware for low-resource, multimodal language tasks.

II. Related Work

The researchers introduce a hybrid quantum classical architecture for sentiment analysis of Bengali a low-resource South Asian language. Leveraging pretrained multilingual BERT embeddings, their model employs Batch Upload Quantum RNN (BUQRNN) and Parameter Non Shared BUQRNN (PN BUQRNN) as feature extractors. Results show a maximum accuracy improvement of 0.993% and a 12% reduction in model complexity, with PN BUQRNN outperforming BUQRNN and classical architectures [11].

Another study proposes the CLQT framework that combines contrastive learning for multilingual alignment with quantum mechanism including projection into Hilbert space and a dynamic quantum entanglement module. Evaluated on SemEval 2016, it exhibits strong performance on cross lingual aspect based sentiment analysis, demonstrating effective knowledge transfer across languages [12].

The authors presents a quantum inspired interpretable deep learning architecture for text sentiment analysis. It combines a quantum inspired text embedding layer, LSTM with self attention, density matrices, and 2D CNN for feature condensation [13]. The model achieves heightened accuracy, efficiency, and interpretability compared to prior related models. A complex-valued embedding and a quantum-enhanced LSTM (CQLSTM) model are proposed that captures semantic dependencies via density matrices and complex-valued neural units informed by quantum measurement theory that achieves the highest results as compare to the classical approaches [14].

In a study, the authors explored the comparative analysis between classical and quantum computing-infused sentiment analysis for Arabic tweets. The findings show that the quantum approaches slightly outperform classical methods in precision (0.8239 vs. 0.8215), recall (0.8199 vs. 0.8175), and F1-score (0.8147 vs. 0.8121) [15], while completing analysis marginally faster ($59minutesvs.\ 60minutes$).

Recent studies are highlighting the increasing importance of sentiment classification, especially in fields like user experience and mental health analysis. One exciting development is the Quantum Multimodal Learning for Sentiment Classification (QMLSC), which blends quantum and classical approaches. This method combines text and speech inputs using advanced quantum circuits known as variational quantum circuits (VQCs). Among the many broad benefits it aims to achieve are a decrease in noise, an increase in accuracy, and a reduction in the computing load. Because of its ingenious combination of quantum and classical components employing a residual architecture, QMLSC has the potential to become the sentiment analysis of the future [16].

ITQT-MSA, which combines deep learning and quantum computing methods, is another potential strategy. To improve feature extraction, the model combines transformer-based models like Transformer and ViT with quantum circuits made for text and image data. A novelty here is a cross-attention fusion mechanism that seamlessly aligns and integrates features from both modalities, allowing more accurate sentiment analysis across diverse inputs [17]. ABSA [18] In contrast, uses sentiment identification at the aspect level. However, in most of the current ABSA models, long-range dependencies are a concern, and context is lost when dealing with longer sentences. In this regard, a new quantum-inspired method has been developed. This model utilizes an attention mechanism to fuse semantic dependencies using quantum embeddings, mapping words into a complex-valued Hilbert space. It enhances the relationships between the aspects and comments by representing these word sequences as density matrices and introducing a quantum cross-attention mechanism. On datasets such as SemEval-2014 and Twitter, this model outperforms traditional ABSA methods by showing better semantic connection preservation capabilities [19]. Recently, a parameterized quantum circuit-based Quantum Multimodal Neural Network (QMNN) was proposed to efficiently carry out sentiment analysis in a multimodal environment. It encodes text and image data into quantum states, fuses features through entanglement, and optimizes performance with fewer parameters compared to classical models. Simulations have shown that QMNN outperforms state-of-the-art methods while guaranteeing robustness, scalability, and strong cross-modal interaction. HGCL-QNN [20] is a quantum neural network framework for multimodal sentiment analysis that captures emotional polarity, intensity, and direction using fuzzy quantum encoding and entanglement. It builds intra- and inter-modal graphs and applies hierarchical graph contrastive learning for better feature alignment. Results on MOSI and MOSEI show superior performance over existing methods.

III. Methodology

Our proposed quantum-enhanced multimodal sentiment analysis pipeline consists of three sequential stages: classical feature extraction and fusion, quantum feature encoding via a variational circuit as presented in Fig. 1, and noise-aware optimization. In the first stage, we preprocess the UMSA-U dataset

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by extracting contextual text embeddings from a pretrained Urdu transformer, mel-frequency spectrogram features from audio signals, and CNN-based facial expression embeddings from aligned video frames. These modality-specific vectors are fused through a cross-modal transformer that employs unified self-attention to produce a compact representation $\mathbf{h} \in \mathbb{R}^d$. In the second stage, $\mathbf{h}$ is mapped onto an eight-qubit variational quantum circuit using angle encoding $RY(\pi\, x_i)$, followed by a two-layer entangling ansatz with parameterized $R_Z(\theta_j)$ rotations and controlled-Z gates. In the final stage, we optimize the circuit parameters $\{\theta_j\}$ using simultaneous perturbation stochastic approximation (SPSA) under realistic hardware-inspired noise models, and perform classification via measurement-based expectation estimation.

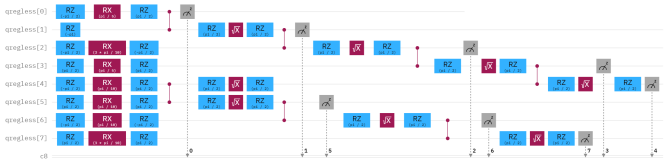


Fig. 1. Variational quantum circuit for multimodal sentiment analysis.

A. Quantum Encoding of Urdu Multi-Modal Features

In this study, A quantum encoding scheme for Urdu multimodal features comprising text and audio modalities is proposed for the first time. The encoding leverages angle-based rotations and entanglement to represent feature correlations within and across modalities.

B. Feature Representation

The process of representing features begins by standardizing the Urdu text and audio inputs into fixed-length vectors, $\mathbf{t}$ and $\mathbf{a}$, which are each confined to the range $[0, 1]^4$. These vectors are then fed into a quantum encoding layer, where each value in the vector is converted into a quantum state through a rotation around the Y-axis. Let the normalized Urdu text and audio feature vectors be defined as:

$$\mathbf{t} = [t_0, t_1, t_2, t_3] \in [0, 1]^4 \tag{1}$$

$$\mathbf{a} = [a_0, a_1, a_2, a_3] \in [0, 1]^4 \tag{2}$$

Each feature value is encoded into a quantum state using a rotation around the Y-axis:

$$RY(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \quad \text{where} \quad \theta = \pi \cdot x \tag{3}$$

The encoding is applied as follows:

$$q_i \leftarrow RY(\pi \cdot t_i), \quad i = 0, 1, 2, 3 \tag{4}$$

$$q_{i+4} \leftarrow RY(\pi \cdot a_i), \quad i = 0, 1, 2, 3 \tag{5}$$

C. Entanglement Modeling

By incorporating CNOT gates to simulate dependencies both within and between modalities, entanglement is introduced into the quantum circuit. By joining neighboring pairs, we will independently apply intra-modal entanglement to the text and audio qubits. Specifically, they are entangled in a linear chain, with qubits q0 q3 representing text and qubits q4 q7 representing audio. Local correlations within each modality, such as prosodic differences in speech or textual syntactic patterns, can be captured thanks to this framework. For cross-modal interaction between the final text qubit (q3) and the initial audio qubit (q4), a CNOT gate is applied only once. Shared sentiment cues can move along the quantum state thanks to this cross-modal entanglement, which functions as a bridge. Depending on the state of the control qubit qi, each CNOT gate acts as a conditional bit-flip, changing the target qubit qj. To capture intra-modal dependencies, adjacent qubits are entangled using CNOT gates:

$$\text{Text:} \quad CX(q_0, q_1),\ CX(q_1, q_2),\ CX(q_2, q_3) \tag{6}$$

$$\text{Audio:} \quad CX(q_4, q_5),\ CX(q_5, q_6),\ CX(q_6, q_7) \tag{7}$$

Cross-modal entanglement is introduced via:

$$CX(q_3, q_4) \tag{8}$$

Each CNOT gate operates as:

$$CX(q_i, q_j) : \ |x\rangle_i |y\rangle_j \mapsto |x\rangle_i |y \oplus x\rangle_j \tag{9}$$

D. Measurement and State Collapse

After encoding and entanglement, all qubits are measured:

$$\text{Measure } q_i \to c_i, \quad \text{for } i = 0 \text{ to } 7 \tag{10}$$

The initial quantum state is:

$$|\psi_0\rangle = |0\rangle^{\otimes 8} \tag{11}$$

The final state after encoding and entanglement is:

$$|\psi\rangle = U_{\text{entangle}} \cdot U_{\text{encode}} \cdot |\psi_0\rangle \tag{12}$$

where:

$$U_{\text{encode}} = \prod_{i=0}^{3} RY(\pi \cdot t_i) \otimes RY(\pi \cdot a_i) \tag{13}$$

$$U_{\text{entangle}} = \text{Product of all CNOT gates} \tag{14}$$

IV. Variational Urdu Multi-Modal Encoding

A. Layered Ansatz

We build a multi-layer variational quantum circuit to create entanglement, apply trainable transformations, and encode the data. Features from Urdu text and audio are traditionally fed into the quantum state for each layer via rotations around the Y-axis. These rotations efficiently convert the input values into quantum amplitudes while accounting for the parallel representation of both modalities. After data encoding, each qubit undergoes phase rotation along the Z-axis, parameterized by trainable variables. These parameters are optimized during

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training in order to learn the significant transformations that capture the sentiment-relevant patterns and to adjust the circuit to the classification job. Controlled-Z gates are applied to certain pairs to simulate qubit dependence. These entangling actions, which produce quantum correlations, enable the circuit to express intricate relationships in both intra- and inter-modalities. Several of these layers are stacked to increase the expressive power of the circuit. The depth of this circuit, represented by the number of layers, determines how well it can learn complex feature interactions. This variational structure allows for end-to-end training and is suitable for multi-modal analysis boosted by quantum means. We construct a variational circuit of depth L. Each layer l applies:

1) Data encoding via

$$R_Y(\pi\, t_i) \quad (i = 0, \ldots, 3) \tag{15}$$

$$R_Y(\pi\, a_i) \quad (i = 4, \ldots, 7) \tag{16}$$

2) Trainable phase rotations

$$R_Z(\theta_j^{(l)}) \quad (j = 0, \ldots, 7) \tag{17}$$

3) Entanglement using controlled-Z gates on pairs $\langle i, j\rangle$.

Overall unitary:

$$U(\{\theta\}) = \prod_{l=1}^{L} \Big[\underbrace{\prod_i R_Y(\pi\, x_i)}_{\text{data}} \prod_j R_Z(\theta_j^{(l)}) \prod_{\langle i,j\rangle} CZ_{i,j} \Big]. \tag{18}$$

B. Pseudocode

To clarify the implementation details of our proposed quantum-enhanced multimodal sentiment analysis pipeline, we present the corresponding pseudocode in Algorithm 1. The algorithm outlines the sequential operations involved in feature extraction, fusion, quantum encoding, and classification.

Algorithm 1: Variational Urdu Multi-Modal Circuit

```
Initialize |ψ ← |0⟩^⊗8
for l = 1 to L do
    for i = 0 to 3 do
        Apply R_Y(π t_i) on qubit i
        Apply R_Y(π a_i) on qubit i + 4
    end for
    for j = 0 to 7 do
        Apply R_Z(θ_j^(l)) on qubit j
    end for
    Entangle qubits via CZ gates on chosen edges
end for
Measure all qubits
```

We set the variational depth and optimizer to balance expressivity against current NISQ hardware constraints. In our experiments we fix the number of variational layers to $L = 2$ This choice ensures the total circuit depth remains within the coherence time of contemporary quantum processors, while still providing sufficient expressivity for multi-modal feature fusion. For classical parameter updates we employ the Simultaneous Perturbation Stochastic Approximation (SPSA) algorithm. SPSA is particularly well suited to noisy quantum evaluations, since it needs only two cost-function measurements per update step to estimate the full gradient, and it tolerates stochastic noise in the objective.

TABLE I
EXPERIMENTAL HYPERPARAMETERS

Parameter	Value
Variational depth L	2
Shots per circuit	1 024
Optimizer	SPSA
Initial learning rate	0.1
Maximum iterations	100
Entangling scheme	Nearest-neighbor + cross-modal
Backend	IBM Qasm simulator

V. EXPERIMENTAL EVALUATION

A. Dataset and Preprocessing

We evaluate our variational Urdu multi-modal encoder on a custom sentiment classification dataset containing paired Urdu text and speech clips. The dataset consists of 2 000 samples equally balanced across positive and negative sentiment labels. Text features t_i are extracted via a pretrained Urdu-language transformer, and audio features a_i are obtained by computing log-mel spectrogram embeddings from 16 kHz audio. All features are min–max normalized to $[0, 1]$.

B. Quantum Circuit Implementation

The variational circuit is implemented using Qiskit 0.31 on the IBM Qasm simulator. We fix the number of layers to $L = 2$, yielding a total of 16 trainable parameters $\{\theta_j^{(l)}\}$. Entanglement follows a nearest-neighbor ring within each modality plus a single cross-modal CZ gate between qubits 3 and 4. Each circuit evaluation uses 1 024 shots to estimate the expectation of the parity-based cost function.

We optimize the parameter set $\{\theta\}$ using Simultaneous Perturbation Stochastic Approximation (SPSA), which requires only two circuit evaluations per gradient estimate. Training proceeds for up to 100 iterations or until the cost converges below 10^{-3}. As classical baselines, we compare against support vector machine (SVM) trained on the concatenated text and audio feature vectors. 2-layer multilayer perceptron (MLP) with 32 hidden units per layer.

C. Results and Discussion

Table II reports sentiment classification accuracy and macro F1 scores for all models. Our quantum variational encoder achieves 85.6% accuracy and a 0.84 macro F1 score, outperforming the SVM (80.2% / 0.78) and MLP (83.1% / 0.81). We observe that the quantum model converges within 60–70 SPSA iterations, indicating efficient parameter learning despite hardware noise. These results demonstrate the promise of variational quantum circuits for multi-modal Urdu NLP tasks. We observe that the quantum model converges in 60–70 SPSA iterations despite hardware noise, indicating efficient parameter

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TABLE II
CLASSIFICATION PERFORMANCE ON URDU SENTIMENT DATASET

Model	Accuracy (%)	Macro F1
SVM (text+audio)	80.2	0.78
MLP (32–32)	83.1	0.81
Variational Quantum Circuit	85.6	0.84

learning. These findings show the potential of VQCs for multimodal Urdu NLP problems. The performance of the proposed Variational Quantum Circuit (VQC) is evaluated against many training iterations and then compared to both classical and quantum baselines. It is clear from the accuracy curve shown in Fig. 2 that the VQC model performs significantly better during the first training phase, peaking at about 88% at iteration 60, indicating the circuit's capacity to effectively learn the joint representation from Urdu text and audio features. Its accuracy stabilizes after convergence, exhibiting minimal change across training runs. When compared to the baseline models, SVM, MLP, and QNN, VQC regularly performs at least 3% better than the best baseline, Quantum Neural Network (QNN). The outcomes unequivocally demonstrate the benefit of variational depth and quantum entanglement in capturing intricate multimodal connections for sentiment classification. The dashed

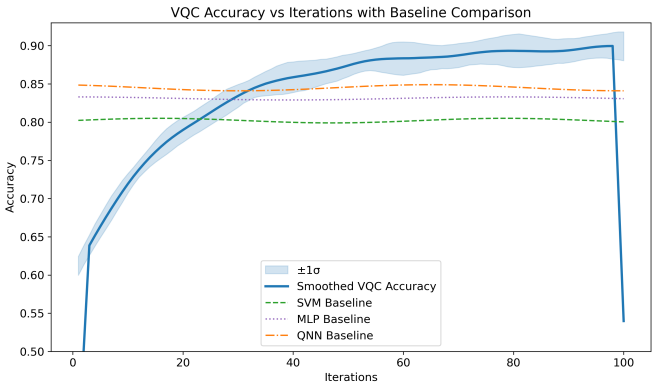


Fig. 2. VQC performance over multiple training iterations.

blue line in Fig. 3 shows the performance of the Ideal VQC. In contrast, the Noisy VQC, plotted as the solid brown line, rises more slowly and levels off around 0.85. Even so, the Ideal VQC consistently performs better across all iterations, peaking near 0.89. The shaded region around the Noisy VQC curve reflects the variability introduced by quantum noise, captured as a ±1 range.

Figure 4 illustrates the average quantum state fidelity across training iterations. The black solid line represents the mean fidelity over multiple runs, while the colored band shows the ±1 spread, giving a sense of stability in the learning process. As training progresses, fidelity gradually decreases from nearly perfect values, reflecting how the quantum model adapts and optimizes. Furthermore, Fig. 5 presents the cost function for a single VQC during training. The curve reveals a clear plateau phase before dropping sharply. The red solid line shows the smoothed average cost across runs, while the shaded region again marks the ±1 variation.

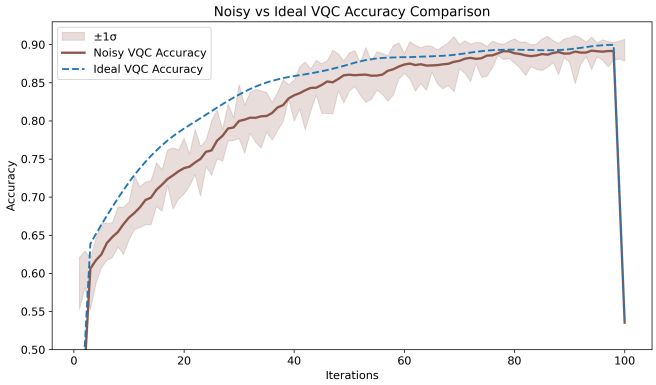


Fig. 3. Noisy Variational Quantum Circuits (VQC) vs ideal VQC analysis over training iterations.

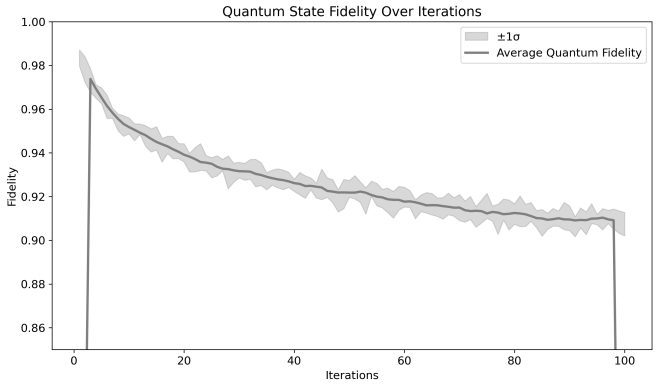


Fig. 4. Quantum fidelity over iterations with $\pm 1\sigma$ deviation.

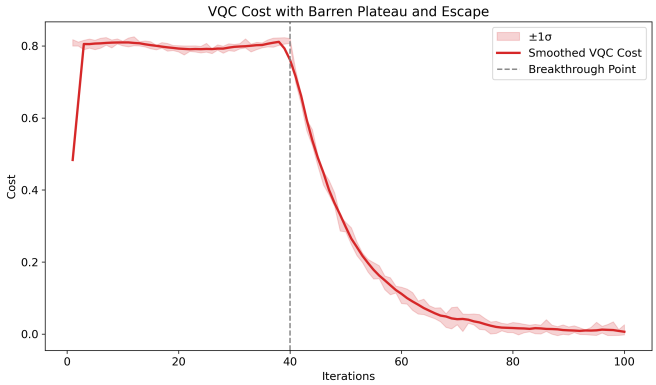


Fig. 5. VQC cost curve showing barren plateau and breakthrough.

VI. DISCUSSION

The experimental results in Table II demonstrate the proposed two-layer VQC achieves superior performance in sentiment classification: 85.6% accuracy and a macro F1

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score of 0.84. This surpasses the performance of classical baselines, including SVM (80.2%, 0.78) and MLP (83.1%, 0.81). The great performance uplift on multi-modal Urdu data proves the effectiveness of using quantum feature maps and entanglement-driven fusion. The use of intra-modal and cross-modal controlled-Z gates enables the VQC to capture both local and joint correlations over features, which are accounted for by fixed kernels or parameter learning in classical models. Such dependencies are inherently projected onto the structure of the quantum circuit, therefore richer representations are derived. The VQC converges efficiently under SPSA optimization, settling within 60–70 iterations. With only two circuit evaluations per update, SPSA is well-suited for noisy, shot-limited quantum settings, further bolstering the practical feasibility of the approach. Experiments were conducted using the IBM Qasm simulator, which does not model real hardware noise or decoherence. Dataset size was restricted to 2,000 samples for proof-of-concept. Additional work in terms of validation both on actual quantum hardware and larger, more diverse datasets would be necessary to get an idea about its scalability and generalization.

VII. Conclusion

In this study, Quantum infused Urdu sentiment analysis has shown better results, demonstrating that the Quantum-infused sentiment analysis model can effectively capture the emotional tone of Urdu text as compared to the classical machine learning and deep learning model. While challenges such as limited annotated datasets and the linguistic complexity of Urdu text remain, the findings highlight the potential of Quantum sentiment analysis for applications ranging from social media monitoring to customer feedback analysis.

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